CariSurg MedTech Pathways Research Project

Preliminary Proposal Template

Project Title: AI-Driven Triage Decision-Support in Emergency Departments: A 12-Week Pilot at Mercer General Hospital

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Statement of the Problem:

Emergency departments (EDs) face persistent overcrowding, inconsistent patient prioritisation and constrained resources. Traditional triage systems, including the Manchester Triage System (MTS), rely on subjective clinician judgment and are particularly vulnerable to error under high-pressure conditions. Recent AI triage models have demonstrated improved predictive performance and operational efficiency; however, the evidence base is limited by poor generalizability, inadequate real-world integration, and insufficient evaluation of patient-centered outcomes, safety (including under-triage risk), and algorithmic bias. This 12-week pilot at Mercer General Hospital will assess an AI triage decision-support tool embedded in routine ED workflow, evaluating triage accuracy, time-to-decision and clinician acceptability

Previous Work:

AI-driven triage in emergency departments: A review of benefits, challenges and future directions

DOI

Emergency Departments face persistent overcrowding, limited resources and inconsistent patient prioritisation due to the subjective nature of traditional triage systems, especially under high-pressure conditions. The authors address this problem through a narrative review of peer-reviewed literature published between 2015 and 2024, sourced from databases including PubMed, Scopus, IEEE Xplore and Google Scholar. The review synthesises findings on AI-driven triage systems that use patient data such as vital signs and clinical information to support automated prioritisation. The reported outcome is that these AI systems generally improve triage accuracy, reduce waiting times, optimise resource allocation and enhance overall emergency department efficiency and patient care support. However, the authors explicitly note limitations, including data quality issues, algorithmic bias, lack of clinician trust and ethical concerns that currently hinder safe and equitable real-world deployment.

Use of Artificial Intelligence in Triage in Hospital Emergency Departments: A Scoping Review

DOI

Emergency departments struggle with triage efficiency, patient prioritization and resource allocation and while AI/ML holds promise in addressing these challenges, their actual impact on ED triage remains poorly characterized in the literature. The authors conducted a scoping review searching EMBASE, Ovid MEDLINE and Web of Science, screening 1,142 citations and selecting 29 peer-reviewed US primary research studies from 2013 - 2023 that used AI during triage and reported patient outcomes. Across those studies, ML models consistently outperformed conventional triage systems in predictive accuracy, disease identification, hospitalization prediction and resource allocation, including length-of-stay forecasting. The authors themselves acknowledge that title/abstract screening relied on subjective judgment, risking selection inconsistencies and that excluding high-bias studies may have omitted valuable insights, limiting the comprehensiveness and generalizability of their conclusions. Notably, the review raises red flags for inflated comparisons by pooling results across heterogeneous cohorts and metrics and never translates performance statistics into bedside consequences - a critical omission given the patient-safety stakes of ED triage.

Clinical Impact of Artificial Intelligence-Based Triage Systems in Emergency Departments: A Systematic Review

DOI

The clinical problem addressed was the growing pressure on emergency departments to improve triage accuracy and efficiency amid increasing patient volumes and limited resources. The method used was a systematic review conducted according to PRISMA 2020 guidelines, analyzing studies from major medical and engineering databases between 2020 and 2025 that evaluated AI-based emergency department triage systems, with risk of bias assessed using an adapted QUADAS-2 tool and findings synthesized narratively. The reported outcome was that AI systems showed promise in improving ED operations, with Voice-AI reducing documentation time by 19% and machine learning models lowering mis-triage rates by 0.3 - 8.9%, while also supporting faster triage and better clinical decision-making. The authors' stated limitations included undertriage risks, inconsistent accuracy across systems, reliance on predominantly single-center studies, workflow integration challenges, limited clinician acceptance data and a lack of evidence regarding patient-centered outcomes, equity and long-term impacts, highlighting the need for multi-center validation and standardized reporting.

Rapid triage for COVID-19 using routine clinical data for patients attending hospital: development and prospective validation of an artificial intelligence screening test

DOI

The clinical problem addressed in this study was the difficulty of rapidly distinguishing COVID-19 from other conditions in emergency department patients, especially given delays and imperfect sensitivity of PCR testing, creating a need for faster screening tools to support triage and patient streaming. The method used was the development and validation of machine learning classifiers (including linear models and gradient boosting methods such as XGBoost) trained on routinely collected electronic health record data-laboratory tests, blood gas measurements, and vital signs-from over 115,000 adult patients across four UK hospitals, with both retrospective evaluation and prospective validation against PCR-confirmed outcomes. The reported outcome was that the AI models achieved strong discriminative performance (AUROC around 0.94), high specificity (about 95%), and very high negative predictive value (>98.5%), enabling effective exclusion of COVID-19 within the first hour of presentation and showing potential to improve patient triage decisions, reduce unnecessary isolation, and support faster clinical workflows. The authors' stated limitations included reliance on a largely UK-based hospital system with limited ethnic diversity, use of pre-pandemic controls which may introduce temporal bias, restriction to adult patients, absence of co-infection data (e.g., influenza and COVID-19), and the need for external international validation to confirm generalizability and real-world robustness.

AI-Based Triage Decision Support: Multisite Economic Evaluation in the United States

DOI

The study addresses the clinical/engineering problem of rising emergency department (ED) crowding and the lack of appropriate economic frameworks to evaluate AI-driven triage tools that aim to improve patient throughput and efficiency. The authors developed and applied a generalizable economic model using operational and financial data from three EDs (170,723 visits over pre- and post-intervention periods) following implementation of an AI triage clinical decision support system, comparing hospital management versus public policy cost-modeling approaches, and conducted sensitivity and break-even analyses across different throughput scenarios. The reported outcome was that AI implementation increased ED revenue by US$15.4 million alongside a 9.6% rise in visit volume, but estimated financial gains varied dramatically depending on the cost framework-yielding a large operating margin gain under the hospital management model (US$12.6 million incremental margin; break-even US$66.02 per visit) versus a minimal gain under the public policy model (US$0.9 million; break-even US$4.69 per visit). The authors' stated limitations include reliance on data from a single health system, potential confounding in attributing volume and length-of-stay changes to the AI system, exclusion of downstream hospital and patient-level financial impacts, assumption of stable costs over time without accounting for learning effects or market changes, and limited generalizability beyond the US healthcare payment context and specific hospital ownership or market conditions, with a call for validation across diverse health systems.

Problem statement

Emergency departments face overcrowding and inconsistent triage decisions driven by subjective scoring systems and variable clinician judgment. Recent AI triage models show improved predictive performance and efficiency. However, evidence is limited by poor generalisability, limited real-world integration and insufficient evaluation of patient-centred outcomes, safety (including under-triage risk) and bias. This

12-week pilot at Mercer General Hospital will assess an AI triage decision-support tool embedded in routine workflow, evaluating triage accuracy, time-to-decision and clinician acceptability.

Proposed Solution:

Deploy a machine-learning-based triage decision-support tool integrated into the existing ED electronic health record (EHR) at Mercer General Hospital. The system will ingest routinely collected patient data at presentation - vital signs, chief complaint, and basic blood results - and generate a real-time acuity score to supplement, not replace, clinician judgement. Over 12 weeks, the pilot will: (1) benchmark AI-generated acuity scores against existing Manchester Triage System (MTS) scores; (2) measure time-to-triage-decision before and after deployment; (3) survey nursing and physician staff on acceptability and trust; and (4) audit under-triage and over-triage rates as primary safety endpoints. All model outputs will be logged for bias auditing across age, sex, and presenting complaint categories.

Gap Analysis

Gap 1
Multiple reviews (Da'Costa et al., 2025; Fernandez Vallejo et al., 2024; Tyler et al., 2024) consistently flag that AI triage models are evaluated on retrospective datasets with minimal evidence of prospective, live deployment. Where real-world pilots exist, clinician acceptance data are sparse or absent. The gap is not that no system has been deployed - it is that deployment studies rarely capture how clinicians interact with AI recommendations under time pressure, whether they override outputs, or how trust evolves across a deployment period. A 12-week Mercer pilot with structured acceptability surveys and override-rate logging could generate the kind of implementation evidence that is systematically absent from the literature.

Gap 2
Fernandez Vallejo et al. (2024) explicitly identify undertriage risk as a limitation insufficiently addressed in existing AI triage studies, while Tyler et al. (2024) note the failure to translate model performance statistics into bedside consequences. Similarly, Da'Costa et al. (2025) call out the lack of patient-centred outcome data. High AUROC scores and reduced waiting times are useful signals, but they do not answer the most important safety question: Does the AI system cause clinicians to miss or delay care for high-acuity patients? A prospective Mercer pilot with pre-specified under-triage and adverse event endpoints would directly address this gap - and would be methodologically stronger than any study currently in the reviewed literature

References :

1. Da'Costa, R., et al. (2025). AI-driven triage in emergency departments: A review of benefits, challenges and future directions.

2. Tyler, P.D., et al. (2024). Use of artificial intelligence in triage in hospital emergency departments: A scoping review.
3. Fernandez Vallejo, A.M., et al. (2024). Clinical impact of artificial intelligence-based triage systems in emergency departments: A systematic review.

4. Soltan, A.A.S., et al. (2021). Rapid triage for COVID-19 using routine clinical data for patients
attending hospital: Development and prospective validation of an artificial intelligence screening test. The Lancet Digital Health.

5. Dong, E., et al. (2024). AI-based triage decision support: Multisite economic evaluation in the United
States.